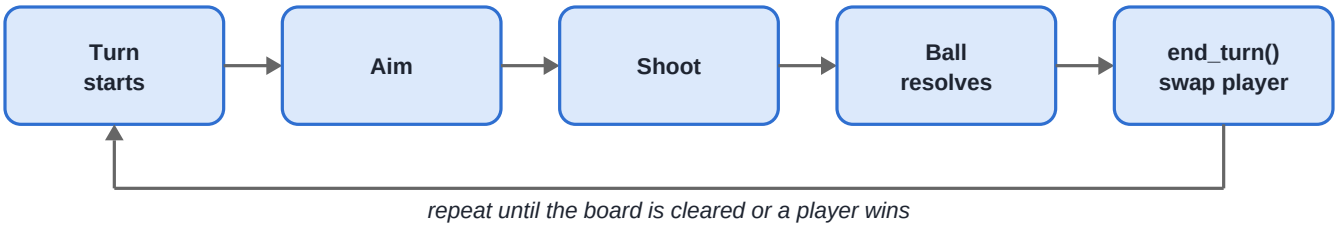
Peggle Battle: build roadmap

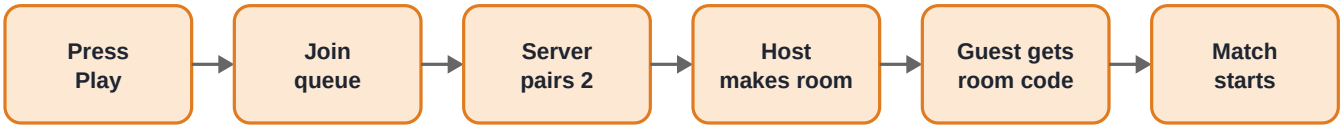
Turn-based, shared board, cross-play PC and mobile, Play-button matchmaking. Build top to bottom. Do not start phase 7 until phase 5 is fun on one device.

How the pieces connect

Turn loop (phase 5, same loop online)



Matchmaking (phase 8)



Who owns what online (phase 7)

Host	Shooter (current turn)	Everyone
Turn order Which pegs are cleared Score and goal count Win check Validates shooter reports	Ball physics while in flight Aim and shoot Reports events: peg hit, caught, lost	Draw the board and effects Send input only on own turn Never change game state directly

Open decisions

- Does the board keep going to full clear after someone claims 5 of 8 goal pegs, or end right there? (You said you dislike early endings.)
- How many balls per turn, and does a bucket catch keep your turn or just add a ball?
- What does the mystery peg give besides a spare ball?

Rules to keep

- Get it playable offline before any networking.
- The turn ends when the ball has fully resolved, not when it is shot.
- Change ball authority only between turns.
- Do not re-run Install Android Build Template, it wipes your swapped encrypted libs. Back them up.

Phase notes

Phase 1: Board and ball

- Playfield fixed at 720x1280 with walls and ceiling, open bottom
- Peg scene (Sprite2D + collision) and ball (RigidBody2D) with gravity and bounce
- Aim with tap/click, shoot from the top (use get_global_mouse_position)
- Stretch: canvas_items + keep aspect, so the board is identical on every device

[ ] Done when: you shoot a ball, it bounces through pegs and falls out the bottom.

Phase 2: Peg logic

- One peg script with a type enum: normal, goal, multiplier, mystery
- Hit pegs light up and clear once the ball finishes its path
- Stuck-ball fix: if the ball barely moves for about 2 seconds, clear the pegs it touches

[ ] Done when: pegs light up, disappear, and the ball can never get stuck forever.

Phase 3: Bucket

- Bucket sweeps left and right (sine or bounce between two limits)
- Catch Area2D inside a U-shaped collision polygon
- Catch gives a free ball; both catch and miss must mark the ball as finished

[ ] Done when: catching and missing both work.

Phase 4: Score and goals

- Score per peg, running multiplier from multiplier pegs
- Per-player goal peg count (5 of 8 wins the board)
- Decide the open question below about boards ending early

[ ] Done when: one player can score, claim goal pegs, and the game knows who is ahead.

Phase 5: Local turns

- Game node with player_left, player_right, current_player, turn_number
- end_turn() is called only when the ball has fully resolved, never on shoot
- Same shared board for both players; pegs cleared by one are gone for the other
- Signals: turn_started and turn_ended, UI listens instead of polling

[ ] Done when: two people can pass one device back and forth and finish a board with a winner.

Phase 6: Mystery buffs

- Start with one buff: spare ball
- Each buff is its own small function so new ones never touch the core loop

[ ] Done when: mystery peg gives a buff and the turn flow still works.

Phase 7: Networking

- Test with two copies of the game on your PC through a room code first
- Host owns turn state, peg state, score, win check
- Shooter owns the ball while it is in flight and reports events (peg hit, caught, lost)
- Host validates: is it that player's turn, does the peg exist, was it already cleared
- Use reliable RPCs for turn changes; change ball authority only between turns

[ ] Done when: two instances play a full match and stay in sync.

Phase 8: Matchmaking

- Small matchmaking server (Steam lobbies will not work, mobile cannot use them)
- Play, join queue, server pairs two players, one hosts a room, other gets the code
- Start simple: first two in the queue match. Skill and trophies come much later
- Handle: cancel while searching, opponent disconnects mid-match, search timeout

[ ] Done when: two devices press Play and land in the same match without typing a code.

Phase 9: Mobile

- Safe areas for notches (DisplayServer.get_display_safe_area())
- Touch aim maps to the same angle as mouse aim
- Test the encrypted Android build on a real phone
- Test PC against phone

[ ] Done when: PC and phone play each other on the same board.

Phase 10: Polish

- More boards, art, sound, hit effects, results screen, rematch
- Later: accounts, ranks, rewards

[ ] Done when: it feels good to play ten matches in a row.

Next action: phase 1. Board, pegs, ball, bucket. No turns, no multiplayer yet.